LOCAL GENERATION, FINITE TYPE, FINITE PRESENTATION, COHERENCE, QUASI-COHERENCE AND LOCAL FREENESS PASS TO TENSOR POWERS, EXTERIOR POWERS AND SYMMETRIC POWERS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The corresponding permanence statements for the tensor powers themselves. Invoked at the start of the counted proof (source lines 3627-3628, 3646-3647) as a pointer to a lemma of this chapter. Reduces all six assertions to statements about the tensor powers, the exterior and symmetric powers being quotients of them.
- The presentation of the symmetric and exterior powers and of their tensor powers by generators and relations, and the behaviour of coherence under exact sequences of modules. Invoked inside the counted proof (source lines 3640-3647, 3650-3657, 3669-3673) as pointers to lemmas of this chapter and to results of another chapter. Produces the presentation used to transfer finite presentation and coherence.

Source reference. Original lemma lemma-tensor-product-permanence: sources/stacks-modules/pinned-original-source.tex, source lines 3628–3628. Navigation pointer only: the counted text refers to the Stacks item lemma-tensor-product-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-presentation-sym-exterior: sources/stacks-modules/pinned-original-source.tex, source lines 3641–3641. Navigation pointer only: the counted text refers to the Stacks item lemma-presentation-sym-exterior, which belongs to another chapter of the project and is not part of this card. The

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Source reference. Original lemma lemma-local-tensor-algebra: sources/stacks-modules/pinned-original-source.tex, source lines 3551–3551. Navigation pointer only: the counted text refers to the Stacks item lemma-local-tensor-algebra, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-coherent-abelian: sources/stacks-modules/pinned-original-source.tex, source lines 2011–2011. Navigation pointer only: the counted text refers to the Stacks item lemma-coherent-abelian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-free-tensor-algebra: sources/stacks-modules/pinned-original-source.tex, source lines 3637–3637. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-free-tensor-algebra, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-present-sym-wedge: sources/stacks-modules/pinned-original-source.tex, source lines 3651–3651. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-present-sym-wedge, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules.*

1. *If $\mathcal{F}$ is locally generated by sections, then so is each $T^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$, and $Sym^n(\mathcal{F})$.*
2. *If $\mathcal{F}$ is of finite type, then so is each $T^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$, and $Sym^n(\mathcal{F})$.*
3. *If $\mathcal{F}$ is of finite presentation, then so is each $T^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$, and $Sym^n(\mathcal{F})$.*
4. *If $\mathcal{F}$ is coherent, then for $n > 0$ each $T^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$, and $Sym^n(\mathcal{F})$ is coherent.*
5. *If $\mathcal{F}$ is quasi-coherent, then so is each $T^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$, and $Sym^n(\mathcal{F})$.*
6. *If $\mathcal{F}$ is locally free, then so is each $T^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$, and $Sym^n(\mathcal{F})$.*

Proof. These statements for $\mathrm{T}^n(\mathcal{F})$ follow from Lemma lemma-tensor-product-permanence.

Statements (1) and (2) follow from the fact that $\wedge^n(\mathcal{F})$ and $\mathrm{Sym}^n(\mathcal{F})$ are quotients of $\mathrm{T}^n(\mathcal{F})$.

Statement (6) follows from Algebra, Lemma algebra-lemma-free-tensor-algebra.

For (3) and (5) we will use Lemma lemma-presentation-sym-exterior above. By locally choosing a presentation $\mathcal{F}_2 \to \mathcal{F}_1 \to \mathcal{F} \to 0$ with $\mathcal{F}_i$ free, or finite free and applying the lemma we see that $\mathrm{Sym}^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$ has a similar presentation; here we use (6) and Lemma lemma-tensor-product-permanence.

To prove (4) we will use Algebra, Lemma algebra-lemma-present-sym-wedge. We may localize on X and assume that $\mathcal{F}$ is generated by a finite set $(s_i)_{i\in I}$ of global sections. The lemma mentioned above combined with Lemma lemma-local-tensor-algebra above implies that for $n \geq 2$ there exists an exact sequence

$$\bigoplus\nolimits_{j\in J} \mathrm{T}^{n-2}(\mathcal{F}) \to \mathrm{T}^n(\mathcal{F}) \to \mathrm{Sym}^n(\mathcal{F}) \to 0$$

where the index set J is finite. Now we know that $\mathrm{T}^{n-2}(\mathcal{F})$ is finitely generated and hence the image of the first arrow is a coherent subsheaf of $\mathrm{T}^n(\mathcal{F})$, see Lemma lemma-coherent-abelian. By that same lemma we conclude that $\mathrm{Sym}^n(\mathcal{F})$ is coherent. $\square$